



S.A.P.D. Jain Pathashala

Walchand Institute of Technology, Solapur

An Autonomous Institute

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Department of Civil Engineering

Lab Details

Lab Name	Fluid Mechanics Lab.
Lab Description	Major tests conducted in the laboratory: 1. Study of fluid pressure and its measurement 2. Calibration of orifice, mouthpiece and Venturimeter calibration 3. Study of flow through pipes and open channels 4. Working of Pelton Wheel and Francis turbines and pumps. 5. Study of Bernoulli's theorem, Reynolds experiment. Study and measurement of discharge through Notches and weirs. Study of Impact of jets. Study of Venturiflumes etc.
Lab Equipment	1. Reynolds Apparatus 2. Losses in Pipe Apparatus 3. Orifice and Mouth Piece Apparatus 4. Notch Apparatus 5. Adjustable Channel (Tilting flume apparatus) 6. Adjustable Channel 7. Water Current Meter 8. Performance Test on Pelton Wheel Turbine test rig 9. Performance test rig on Francis turbine test rig 10. Francis turbine. 11. Pelton wheel turbine 12. Channel 0.45m wide and 80m length 13. Adjustable channel and accessories 14. R.C.C. constant head tank (12m) 15. Impact of jet apparatus and accessories. 16. Losses in pipelines apparatus and accessories. 17. Notch Tank apparatus and accessories 18. Orifice and mouthpiece apparatus and accessories 19. Pumps and valves apparatus dismantling and refitting 20. Bernoulli's Apparatus
Lab Floor Area in square meters	174
Practicals Engaged in Lab	For semester I 1. Study of manometers 2. Measurement of pressure and discharge



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	3. Determination of Metacentric height 4. Verification of Bernoulli's theorem 5. Calibration of Venturi meter 6. Calibration of Orifices 7. Calibration of Mouthpieces 8. Study of losses in pipes 9. Determination of Darcy's factor 10. Verification of streamlines and flow nets 11. Study of Moody's charts and Nomographs For semester II 1. Calibration of Broad crested weir 2. Calibration of Sharp crested weir 3. Determination of coefficient of discharge of Ogee spillway 4. Determination of coefficient of discharge of Venturi flume 5. To calculate critical velocity by Reynold's apparatus 6. Study of Hydraulic jump 7. Study of the Impact of jets 8. Study of Pelton wheel 9. Study of Francis turbine 10. Study of properties of fluids 11. 11. Classification and silent features of Turbo-machines
Name of Lab Incharge	Dr. Deshmukh Sachin Chandrakant
Associated Faculty with the lab	Dr. Satishkumar Shahajirao Kashid
Name of Lab Assistant	Mr. S. K. Umbardand
Name/s of Menial Staff	Mr. Ashok Dhotre